Database System Principle - SQL Query Language

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OBJECTIVES

- Overview of The SQL Query Language
- Data Definition
- Basic Query Structure
- Additional Basic Operations
- Set Operations
- Null Values
- Aggregate Functions
- Nested Subqueries
- Modification of the Database

OVERVIEW OF THE SQL QUERY LANGUAGE

- IBM **Sequel** language developed as part of System R project at the IBM San Jose Research Laboratory
  - 1970s
  - Renamed Structured Query Language (SQL)
  - In 1986, the American National Standards Institute (ANSI) and the International Organization for Standardization (ISO) published an SQL standard, called SQL-86

ANSI AND ISO STANDARD SQL

- SQL-86 (or SQL-87) is the ISO 9075:1987 standard of 1987
- SQL-89 is the ISO/IEC 9075:1989 standard of 1989
- SQL-92 is the ISO/IEC 9075:1992 standard of 1992
- SQL:1999 is the ISO/IEC 9075:1999 standard of 1999
- SQL:2003 is the ISO/IEC 9075:2003 standard of 2003*
- SQL:2006 is the ISO/IEC 9075:2006 standard of 2006
- SQL:2008 is the ISO/IEC 9075:2008 standard of 2008
- SQL:2011 is the ISO/IEC 9075:2011 standard of 2011

SQL2003

- 1) ISO/IEC 9075—1: Framework(SQL/Framework)
- 2) ISO/IEC 9075—2: Foundation(SQL/Foundation)
- 3) ISO/IEC 9075—3: Call Level Interface(SQL/CLI)
- 4) ISO/IEC 9075—4: Persistent Stored Modules(SQL/PSM)
- 5) ISO/IEC 9075—9: Management of External Data(SQL/MED)
- 6) ISO/IEC 9075—10: Object Language Bindings(SQL/OLB)
- 7) ISO/IEC 9075—11: Information and Definition Schemas(SQL/Schemata)
- 8) ISO/IEC 9075—13: Java Routines and Types Using the Java Programming Language(SQL/JRT)
- 9) ISO/IEC 9075—14: XML Related Specifications(SQL/XML)

OVERVIEW OF THE SQL QUERY LANGUAGE

- Structured Query Language (SQL)
  - Data-Definition Language, DDL
  - Data-Manipulation Language, DML
  - Integrity完整性
  - View Definition视图定义
  - Transaction Control事务控制
  - Embedded SQL and dynamic SQL嵌入式和动态SQL
  - Authorization授权
- Commercial systems offer most, if not all, SQL-92 features, plus varying feature sets from later standards and special proprietary features.

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SQL DATA DEFINITION

- SQL DDL
  - The schema for each relation
  - The types of values associated with each attribute
  - The integrity constraints
  - The set of indices to be maintained for each relation
  - The security and authorization information for each relation
  - The physical storage structure of each relation on disk

DOMAIN TYPES IN SQL

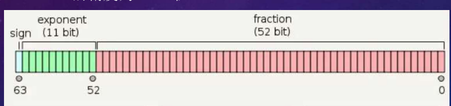
- char(n). Fixed length character string, with user-specified length n.
- varchar(n). Variable length character strings, with user-specified maximum length n.

DOMAIN TYPES IN SQL

- int. Integer (a finite subset of the integers that is machine-dependent).
- smallint. Small integer (a machine-dependent subset of the integer domain type).
- numeric(p,d). Fixed point number, with user-specified precision of p digits, with d digits to the right of decimal point.
  - (ex., numeric(3,1), allows 44.5, not 444.5 or 0.32)
- ...

DOMAIN TYPES IN SQL

- real, double precision. Floating point浮点 and double-precision双精度 floating point numbers, with *machine-dependent precision*
  - 1 bit (sign bit),11 bits (exponential bit), 52 bits (tail digit bit), 2^52 = 4503599627370496
  - 16bits，double的精度为15~16位



- float(n). 单精度 Floating point number, with user-specified precision of at least n digits
  - 1 bit (sign bit) 8 bits (exponential指数 bit) 23 bits (tail digit bit), 2^23 = 8388608, 7bits, float的精度为6~7位有效数字

DATA TYPE IN MYSQL

- Numeric Data Types
  - Integer Types (Exact Value)
    - INTEGER, INT, SMALLINT, TINYINT, MEDIUMINT, **BIGINT**
  - Fixed-Point Types (Exact Value)
    - DECIMAL十进制的，NUMERIC
  - Floating-Point Types (Approximate Value)
    - FLOAT, DOUBLE
  - Bit-Value Type - BITs

DATA TYPE IN MYSQL

- Date and Time Data Types
  - The DATE Type: '1000-01-01' to '9999-12-31'
  - The DATETIME Type: 'YYYY-MM-DD hh:mm:ss', 'YYYY-MM-DD hh:mm:ss[.fraction]'
    - '1000-01-01 00:00:00.000000' to '9999-12-31 23:59:59.499999'
  - The TIMESTAMP Type: 'YYYY-MM-DD hh:mm:ss[.fraction]'
    - '1970-01-01 00:00:01.000000' to '2038-01-19 03:14:07.499999'
  - The TIME Type: 'hh:mm:ss': '-838:59:59' to '838:59:59'
  - The YEAR Type: YYYY format, with a range of 1901 to 2155, and 0000
  - Automatic Initialization and Updating for TIMESTAMP and DATETIME

DATA TYPE IN MYSQL

- String Data Types
  - The CHAR and VARCHAR Types
  - The BINARY and VARBINARY Types
  - The BLOB and TEXT Types
  - The ENUM Type
    - ENUM('x-small', 'small', 'medium', 'large', 'x-large')
  - The SET Type

BASIC SCHEMA DEFINITION

- Create table
- Insert into
- Delete from
- Alter table
- Drop table

CREATE TABLE CONSTRUCT

- An SQL relation is defined using the **create table** command:

  create table *r* (A_1 D_1,
  A_2 D_2, ...,
  A_n D_n,
  (integrity-constraint$_1$完整性约束),
  ...,
  (integrity-constraint$_k$))

  - *r* is the name of the relation
  - each A_i is an attribute name in the schema of relation *r*
  - D_i is the data type of values in the domain of attribute A_i

CREATE TABLE CONSTRUCT

- the **create table** command
- Example:

create table *instructor* (
ID **char**(5),
name **varchar**(20),
dept_name **varchar**(20),
salary **numeric**(8,2));

INTEGRITY CONSTRAINTS IN CREATE TABLE

- **not null**
- **primary key** (A_1, ..., A_n)
- **foreign key** (A_m, ..., A_n) **references** *r*

Example:

create table *instructor* (
ID **char**(5),
name **varchar**(20) **not null,**
dept_name **varchar**(20),
salary **numeric**(8,2),
primary key (*ID*),
foreign key (*dept_name*) **references** *department);*

primary key declaration on an attribute automatically ensures **not null**

CASES: CREATE TABLE

create table department
(dept_name varchar (20),
building varchar (15),
budget numeric (12,2),
primary key (dept_name);

create table course
(course_id varchar (7),
title varchar (50),
dept_name varchar (20),
credits numeric (2,0),
primary key (course_id),
foreign key (dept_name) references department(dept_name));

CASES: CREATE TABLE

create table section
(course_id varchar (8),
sec_id varchar (8),
semester varchar (6),
year numeric (4,0),
building varchar (15),
room_number varchar (7),
time_slot_id varchar (4),
primary key (course_id, sec_id, semester, year),
foreign key (course_id) references course);

CASES: CREATE TABLE

create table teaches
(ID varchar (5),
course_id varchar (8),
sec_id varchar (8),
semester varchar (6),
year numeric (4,0),
primary key (ID , course_id, sec_id, semester, year),
foreign key (course_id, sec_id, semester, year) references section,
foreign key (ID) references instructor);

UPDATES TO TABLES

- **Insert**
  - **insert into** *instructor* **values** ('10211', 'Smith', 'Biology', 66000);
- **Delete**
  - Remove all tuples from the *student* relation
    - **delete from** *student*
- **Drop Table**
  - **drop table** *r*

UPDATES TO TABLES

- **Alter**
  - **alter table** *r* **add** *A D*
    - where *A* is the name of the attribute to be added to relation *r* and *D* is the domain of *A.*
    - All exiting tuples in the relation are assigned *null* as the value for the new attribute.
  - **alter table** *r* **drop** *A*
    - where *A* is the name of an attribute of relation *r*
    - Dropping of attributes not supported by many databases.

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BASIC STRUCTURE OF SQL QUERIES

- A typical SQL query has the form:

  select $A_1, A_2, ..., A_n$
  from $r_1, r_2, ..., r_m$
  where P

  A_i represents an attribute
  - R_i represents a relation
  - P is a predicate.
- The result of an SQL query is a relation.

QUERIES ON A SINGLE RELATION

- The **select** clause lists the attributes desired in the result of a query
  - corresponds to the projection operation of the relational algebra
- Example: find the names of all instructors:

  select *name*
  from *instructor*
- NOTE: SQL names are case insensitive
  - E.g., *Name* ≡ *NAME* ≡ *name*
  - Some people use upper case wherever we use bold font.

name
Srinivasan
Wu
Mozart
Einstein
El Said
Gold
Katz
Califieri
Singh
Crick
Brandt
Kim

QUERIES ON A SINGLE RELATION - DEDUPLICATION去重

- SQL allows duplicates in relations as well as in query results.
- To force the elimination of duplicates, insert the keyword distinct after select**.**
- Find the department names of all instructors, and remove duplicates

  select distinct *dept_name*
  from *instructor*
- The keyword **all** specifies that duplicates should not be removed.

  select all *dept_name*
  from *instructor*

QUERIES ON A SINGLE RELATION - MORE

- An asterisk (*) in the select clause denotes "all attributes"

  select *
  from *instructor*
- An attribute can be a literal with no **from** clause

  select '437'
  - Results is a table with one column and a single row with value "437"
  - Can give the column a name using:

    select '437' **as** *FOO*

QUERIES ON A SINGLE RELATION - MORE

- An attribute can be a literal with **from** clause

  select 'A'
  from *instructor*
  - Result is a table with one column and N rows (number of tuples in the *instructors* table), each row with value "A"

QUERIES ON A SINGLE RELATION - MORE

- The select clause can contain arithmetic expressions involving the operation, +, –, ∗, and /, and operating on constants or attributes of tuples.
  - The query:

    select *ID, name, salary/12*
    from *instructor*

    would return a relation that is the same as the *instructor* relation, except that the value of the attribute *salary* is divided by 12.
  - Can rename *"salary/12"* using the **as** clause:

    select *ID, name, salary/12* as *monthly_salary*

THE WHERE CLAUSE

- The **where** clause specifies conditions that the result must satisfy
  - Corresponds to the selection predicate of the relational algebra.
- To find all instructors in Comp. Sci. dept

  select *name*
  from *instructor*
  where *dept_name* = 'Comp. Sci.'

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THE WHERE CLAUSE

- Comparison results can be combined using the logical connectives **and**, **or**, and **not**
  - To find all instructors in Comp. Sci. dept with salary > 80000

    select *name*
    from *instructor*
    where *dept_name* = 'Comp. Sci.' **and** *salary* > 80000
- Comparisons can be applied to results of arithmetic expressions.

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QUERIES ON MULTIPLE RELATIONS

select A_1 , A_2 ,..., A_n

from r_1 , r_2 ,...,r_m

where P;

- Each A_i represents an attribute, and each r_i a relation.
- P is a predicate
  - If the where clause is omitted, the predicate P is true.

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QUERIES ON MULTIPLE RELATIONS

- The **from** clause lists the relations involved in the query
  - Corresponds to the Cartesian product operation of the relational algebra.
- Find the Cartesian product *instructor X teaches*

  select *
  from *instructor, teaches*
  - generates every possible instructor – teaches pair, with all attributes from both relations.
  - For common attributes (e.g., *ID*), the attributes in the resulting table are renamed using the relation name (e.g., *instructor.ID*)

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CARTESIAN PRODUCT 笛卡尔积

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000

ID	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009

Inst.ID	*name*	*dept_name*	*salary*	*teaches.ID*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000	10101	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000	10101	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000	10101	CS-347	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000	12121	FIN-201	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000	15151	MU-199	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
12121	Wu	Finance	90000	10101	CS-101	1	Fall	2009
12121	Wu	Finance	90000	10101	CS-315	1	Spring	2010
12121	Wu	Pinance	90000	10101	CS-347	1	Fall	2009
12121	Wu	Pinance	90000	12121	FIN-201	1	Spring	2010
12121	Wu	Finance	90000	15151	MU-199	1	Spring	2010
12121	Wu	Pinance	90000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...

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CASES: CARTESIAN PRODUCT

- Find the names of all instructors who have taught some course and the course_id
  - **select** *name, course_id*
    from *instructor , teaches*
    where *instructor.ID = teaches.ID*
- Find the names of all instructors in the Art department who have taught some course and the course_id
  - **select** *name, course_id*
    from *instructor , teaches*
    where *instructor.ID = teaches.ID* ***and*** *instructor. dept_name* = 'Art'

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THE NATURAL JOIN

select $A_1, A_2, \ldots, A_n$
from r_1 **natural join** r_2 **natural join** ... **natural join** r_m
where P;

More generally, a **from** clause can be of the form

from $E_1, E_2, \ldots, E_n$

- where each Ei can be a single relation or an expression involving natural joins.

CASES: THE NATURAL JOIN

The natural join of the instructor relation with the teaches relation

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000

ID	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009

ID	*name*	*dept_name*	*salary*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000	CS-347	1	Fall	2009
12121	Wu	Finance	90000	FIN-201	1	Spring	2010
15151	Mozart	Music	40000	MU-199	1	Spring	2010
22222	Einstein	Physics	95000	PHY-101	1	Fall	2009
32343	El Said	History	60000	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000	CS-319	1	Spring	2010
76766	Crick	Biology	72000	BIO-101	1	Summer	2009
76766	Crick	Biology	72000	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000	EE-181	1	Spring	2009

CASES: THE NATURAL JOIN

- Way 1

select name, course_id
from instructor, teaches
where instructor.ID = teaches.ID;

- Way 2

select name, course_id
from instructor natural join teaches;

CASES 2: THE NATURAL JOIN

1. **select** *name*, *title*
from *instructor* **natural join** *teaches*, *course*
where *teaches.course_id*= *course.course_id*;

2. **select** *name*, *title*
from *instructor* **natural join** *teaches* **natural join** *course*;

3. **select** *name*, *title*
from (*instructor* **natural join** *teaches*) **join** *course* **using** (*course_id*);

CASES 2: THE NATURAL JOIN -> SCHEMA OF THE UNIVERSITY DATABASE

classroom(*building*, *room_number*, *capacity*)
department(*dept_name*, *building*, *budget*)
course(*course_id*, *title*, *dept_name*, *credits*)
instructor(*ID*, *name*, *dept_name*, *salary*)
section(*course_id*, *sec_id*, *semester*, *year*, *building*, *room_number*, *time_slot_id*)
teaches(*ID*, *course_id*, *sec_id*, *semester*, *year*)
student(*ID*, *name*, *dept_name*, *tot_cred*)
takes(*ID*, *course_id*, *sec_id*, *semester*, *year*, *grade*)
advisor(*s_ID*, *i_ID*)
time_slot(*time_slot_id*, *day*, *start_time*, *end_time*)
prereq(*course_id*, *prereq_id*)

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THE RENAME OPERATION

- The SQL allows renaming relations and attributes using the **as** clause:

  old-name **as** *new-name*

- Find the names of all instructors who have a higher salary than some instructor in 'Comp. Sci'.
  - **select distinct** *T.name*
    from *instructor* **as** *T, instructor* **as** *S*
    where *T.salary > S.salary* **and** *S.dept_name = 'Comp. Sci.'*
- Keyword **as** is optional and may be omitted

  instructor **as** *T ≡ instructor T*

STRING OPERATIONS

- SQL includes a string-matching operator for comparisons on character strings. The operator **like** uses patterns that are described using two special characters:
  - percent (%). The % character matches any substring.
  - underscore (_). The _ character matches any character.

STRING OPERATIONS

- Find the names of all instructors whose name includes the substring "dar".

  select *name*
  from *instructor*
  where *name* **like** '%dar%'

- Match the string "100%"

  like '100 \%' **escape** '\'

  in that above we use backslash (\) as the escape character.

STRING OPERATIONS

- Patterns are case sensitive.
- Pattern matching examples:
  - 'Intro%' matches any string beginning with "Intro".
  - '%Comp%' matches any string containing "Comp" as a substring.
  - '_ _ _' matches any string of exactly three characters.
  - '_ _ _ %' matches any string of at least three characters.

https://dev.mysql.com/doc/refman/8.0/en/string-functions.html

STRING OPERATIONS

- SQL supports a variety of string operations such as
  - concatenation (using "||")
  - converting from upper to lower case (and vice versa)
    - lower(), upper()
  - finding string length, extracting substrings, etc.
    - LENGTH()
    - CHAR_LENGTH()
    - Substr(str,pos[,length])

ORDERING THE DISPLAY OF TUPLES

- List in alphabetic order the names of all instructors

  select distinct *name*
  from *instructor*
  order by *name*

- We may specify **desc** for descending order or **asc** for ascending order, for each attribute; ascending order is the default.
  - Example: **order by** *name* **desc**
- Can sort on multiple attributes
  - Example: **order by** *dept_name, name*

WHERE CLAUSE PREDICATES

- SQL includes a **between** comparison operator
- Example: Find the names of all instructors with salary between \$90,000 and \$100,000 (that is, ≥ \$90,000 and ≤ \$100,000)
  - **select** *name*
    from *instructor*
    where *salary* **between** 90000 **and** 100000

WHERE CLAUSE PREDICATES

- Way 1:
  - **select** *name*, *course_id*
    from *instructor*, *teaches*
    where *instructor.ID= teaches.ID and dept_name* = 'Biology';
- Way 2:
  - **select** *name*, *course_id*
    from *instructor*, *teaches*
    where (*instructor.ID*, *dept_name*) = (*teaches.ID*, 'Biology');

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SET OPERATIONS

- union
- intersect
- except

SET OPERATIONS
- UNION

- Find courses that ran in Fall 2009 or in Spring 2010

(**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
union
(**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

SET OPERATIONS
- INTERSECT

- Find courses that ran in Fall 2009 and in Spring 2010

(**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
intersect
(**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

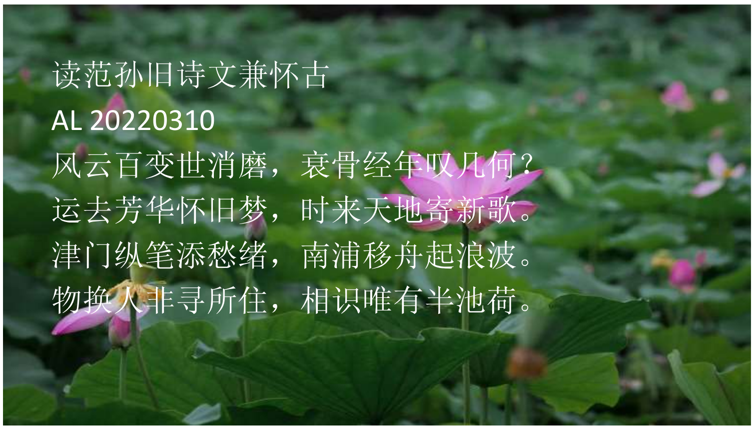
SET OPERATIONS
- EXCEPT

- Find courses that ran in Fall 2009 but not in Spring 2010

(**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
except
(**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

SET OPERATIONS

- Set operations **union**, **intersect**, and **except**
  - Each of the above operations automatically eliminates duplicates
- To retain all duplicates use the corresponding multiset versions **union all**, **intersect all** and **except all**.
- Suppose a tuple occurs *m* times in *r* and *n* times in *s*, then, it occurs:
  - $m + n$ times in *r* **union all** *s*
  - min(*m,n)* times in *r* **intersect all** *s*
  - max(0, $m - n$) times in *r* **except all** *s*



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NULL VALUES

- It is possible for tuples to have a null value, denoted by *null*, for some of their attributes
- *null* signifies an unknown value or that a value does not exist.
- The result of any arithmetic expression involving *null* is *null*
  - Example: 5 + *null* returns null
- The predicate **is null** can be used to check for null values.
  - Example: Find all instructors whose salary is null.

select *name*
from *instructor*
where *salary* **is null**

NULL VALUES AND THREE VALUED LOGIC

- Three logic values – *true*, *false*, *unknown*
- Any comparison with *null* returns *unknown*
  - Example: *5 < null or null <> null or null = null*

NULL VALUES AND THREE VALUED LOGIC

- Three-valued logic using the value *unknown*:
  - OR: (*unknown* **or** *true*) = *true*,
    (*unknown* **or** *false*) = *unknown*
    (*unknown* **or** *unknown*) = *unknown*
  - AND: (*true* **and** *unknown*) = *unknown*,
    (*false* **and** *unknown*) = *false*,
    (*unknown* **and** *unknown*) = *unknown*
  - NOT: (**not** *unknown*) = *unknown*
  - "*P* **is unknown**" evaluates to true if predicate *P* evaluates to *unknown*
- Result of **where** clause predicate is treated as *false* if it evaluates to *unknown*

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AGGREGATE FUNCTIONS聚集函数

- These functions operate on the multiset of values of a column of a relation, and return a value

avg: average value
min: minimum value
max: maximum value
sum: sum of values
count: number of values

AGGREGATE FUNCTIONS

- Find the average salary of instructors in the Computer Science department
  - **select avg** (*salary*)
    from *instructor*
    where *dept_name*= 'Comp. Sci.';
- Find the total number of instructors who teach a course in the Spring 2010 semester
  - **select count** (**distinct** *ID*)
    from *teaches*
    where *semester* = 'Spring' **and** *year* = 2010;

AGGREGATE FUNCTIONS – GROUP BY

- Find the average salary of instructors in each department
  - **select** *dept_name*, **avg** (*salary*) **as** *avg_salary*
    from *instructor*
    group by *dept_name*;

ID	*name*	*dept_name*	*salary*
76766	Crick	Biology	72000
45565	Katz	Comp. Sci.	75000
10101	Srinivasan	Comp. Sci.	65000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
22222	Einstein	Physics	95000

dept_name	*salary*
Biology	72000
Comp. Sci.	77333
Elec. Eng.	80000
Finance	85000
History	61000
Music	40000
Physics	91000

AGGREGATE FUNCTIONS – GROUP BY

- Attributes in **select** clause outside of aggregate functions must appear in **group by** list
  - /* erroneous query */
    select *dept_name*, *ID*, **avg** (*salary*)
    from *instructor*
    group by *dept_name*;

AGGREGATE FUNCTIONS – HAVING CLAUSE

- Find the names and average salaries of all departments whose average salary is greater than 42000

select *dept_name*, **avg** (*salary*)
from *instructor*
group by *dept_name*
having avg (*salary*) > 42000;

Note: predicates in the **having** clause are applied after the formation of groups whereas predicates in the **where** clause are applied before forming groups

AGGREGATE FUNCTIONS

- Total all salaries

select sum (*salary*)
from *instructor*

  - Above statement ignores null amounts
  - Result is *null* if there is no non-null amount

AGGREGATE FUNCTIONS

- All aggregate operations except **count(*)** ignore tuples with null values on the aggregated attributes
- What if collection has only null values?
  - count returns 0
  - all other aggregates return null

OBJECTIVES

- Overview of The SQL Query Language
- Data Definition
- Basic Query Structure
- Additional Basic Operations
- Set Operations
- Null Values
- Aggregate Functions
- Nested Subqueries
- Modification of the Database

NESTED SUBQUERIES嵌套子查询

- SQL provides a mechanism for nesting subqueries.
- A subquery is a select-from-where expression that is nested within another query.

select distinct *course_id*
from *section*
where *semester* = 'Fall' **and** *year*= 2009 **and**
course_id **in** (**select** *course_id*
from *section*
where *semester* = 'Spring' **and** *year*= 2010);

NESTED SUBQUERIES嵌套子查询

- The nesting can be done in the following SQL query

select $A_1, A_2, ..., A_n$
from $r_1, r_2, ..., r_m$
where P

as follows:

  - A_i can be replaced be a subquery that generates a single value.
  - r_i can be replaced by any valid subquery
  - P can be replaced with an expression of the form:

    B <operation> (subquery)

  Where B is an attribute and <operation> to be defined later.

SUBQUERIES IN THE WHERE CLAUSE

- A common use of subqueries is to perform tests:
  - For set membership
  - For set comparisons
  - For set cardinality基数

NESTED SUBQUERIES - SET MEMBERSHIP集合成员测试

- in, not in

 Find courses offered in Fall 2009 but not in Spring 2010

```
select distinct course_id
from section
where semester = 'Fall' and year= 2009 and
      course_id  not in (select course_id
                         from section
                         where semester = 'Spring' and year= 2010);
```


NESTED SUBQUERIES - SET MEMBERSHIP

- Find the total number of (distinct) students who have taken course sections taught by the instructor with *ID* 10101

```
select count (distinct ID)
from takes
where (course_id, sec_id, semester, year) in
                    (select course_id, sec_id, semester, year
                     from teaches
                     where teaches.ID= 10101);
```


NESTED SUBQUERIES - SET COMPARISON集合比较

- Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

```
select distinct T.name
from instructor as T, instructor as S
where T.salary > S.salary and S.dept_name = 'Biology';
```

 Same query using > some clause

```
select name
from instructor
where salary > some (select salary
                     from instructor
                     where dept_name = 'Biology');
```


NESTED SUBQUERIES - SET COMPARISON集合比较

- > some
- < some
- <= some
- >= some
- <> some
  - (not same to not in)
- =some (same to in)

- F <comp> **some** $r \Leftrightarrow \exists\, t \in r$ such that (F <comp> t)
  Where <comp> can be: $<, \leq, >, =, \neq$

(5 < **some** [0, 5, 6]) = true (read: 5 < some tuple in the relation)

(5 < **some** [0, 5]) = false

(5 = **some** [0, 5]) = true

(5 $\neq$ **some** [0, 5]) = true (since 0 $\neq$ 5)

(= **some**) $\equiv$ **in**

However, ($\neq$ **some**) $\not\equiv$ **not in**

NESTED SUBQUERIES - SET COMPARISON集合比较

- > all
- < all
- <=all
- >=all
- =all (not same to in)
- <> all (same to not in)

- F <comp> **all** $r \Leftrightarrow \forall\, t \in r$ (F <comp> t)

(5 < **all** [0, 5, 6]) = false

(5 < **all** [6, 10]) = true

(5 = **all** [4, 5]) = false

(5 $\neq$ **all** [4, 6]) = true (since 5 $\neq$ 4 and 5 $\neq$ 6)

($\neq$ **all**) $\equiv$ **not in**

However, (= **all**) $\not\equiv$ **in**

NESTED SUBQUERIES
- SET COMPARISON集合比较

?What meaning?

```
select dept_name
from instructor
group by dept_name
having avg (salary) >= all (select avg (salary)
                            from instructor
                            group by dept_name);
```


TEST FOR EMPTY RELATIONS

- exists , not exists
- Yet another way of specifying the query "Find all courses taught in both the Fall 2009 semester and in the Spring 2010 semester"

```
select course_id
from section as S
where semester = 'Fall' and year = 2009 and
      exists (select *
              from section as T
              where semester = 'Spring' and year= 2010 and S.course_id = T.course_id);
```

- Correlation name – variable S in the outer query
- Correlated subquery – the inner query

TEST FOR EMPTY RELATIONS

- Find all students who have taken all courses offered in the Biology department.

```
select distinct S.ID, S.name
from student as S
where not exists ( (select course_id   from course
                    where dept_name = 'Biology')
                   except
                   (select T.course_id   from takes as T
                    where S.ID = T.ID));
```

- First nested query lists all courses offered in Biology
- Second nested query lists all courses a particular student took
- Note that $X - Y = \emptyset \Leftrightarrow X \subseteq Y$
- *Note:* Cannot write this query using = **all** and its variants

TEST FOR ABSENCE OF DUPLICATE TUPLES

- The **unique** construct tests whether a subquery has any duplicate tuples in its result.
- The **unique** construct evaluates to "true" if a given subquery contains no duplicates .
- Find all courses that were offered at most once in 2009

```
select T.course_id
from course as T
where unique (select R.course_id
              from section as R
              where T.course_id= R.course_id
                and R.year = 2009);
```


SUBQUERIES IN THE FROM CLAUSE

- SQL allows a subquery expression to be used in the **from** clause
- Find the average instructors' salaries of those departments where the average salary is greater than $42,000."

```
select dept_name, avg_salary
from (select dept_name, avg (salary) as avg_salary
      from instructor
      group by dept_name)
where avg_salary > 42000;
```

- Note that we do not need to use the **having** clause

SUBQUERIES IN THE FROM CLAUSE

- Find the average instructors' salaries of those departments where the average salary is greater than $42,000."
- Another way to write above query

```
select dept_name, avg_salary
from (select dept_name, avg (salary)
      from instructor
      group by dept_name) as dept_avg (dept_name, avg_salary)
where avg_salary > 42000;
```

WITH CLAUSE

- The **with** clause provides a way of defining a temporary relation whose definition is available only to the query in which the **with** clause occurs
  - Introduced in SQL:1999
- Find all departments with the maximum budget

```
with max_budget (value) as
    (select max(budget)
     from department)
select department.name
from department, max_budget
where department.budget = max_budget.value;
```

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COMPLEX QUERIES USING WITH CLAUSE

- Find all departments where the total salary is greater than the average of the total salary at all departments

```
with dept _total (dept_name, value) as
    (select dept_name, sum(salary)
     from instructor
     group by dept_name),
dept_total_avg(value) as
    (select avg(value)
     from dept_total)
select dept_name
from dept_total, dept_total_avg
where dept_total.value > dept_total_avg.value;
```

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SCALAR SUBQUERIES标量子查询

- Scalar subqueries
  - SQL allows subqueries to occur wherever an expression returning a value is permitted, provided the subquery returns only one tuple containing a single attribute
- List all departments along with the number of instructors in each department

```
select dept_name,
        (select count(*)
         from instructor  where department.dept_name = instructor.dept_name)
        as num_instructors
  from department;
```

- Runtime error if subquery returns more than one result tuple

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MODIFICATION OF THE DATABASE

- Deletion of tuples from a given relation.
- Insertion of new tuples into a given relation
- Updating of values in some tuples in a given relation

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DELETION删除

- Delete all instructors

```
delete from instructor
```

- Delete all instructors from the Finance department

```
delete from instructor
where dept_name= 'Finance';
```

- Delete all tuples in the *instructor* relation for those instructors associated with a department located in the Watson building.

```
delete from instructor
where dept name in (select dept name
                    from department
                    where building = 'Watson');
```

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DELETION

- Delete all instructors whose salary is less than the average salary of instructors

```
delete from instructor
where salary < (select avg (salary)
                from instructor);
```

- Problem: as we delete tuples from deposit, the average salary changes
- Solution used in SQL:
  1. First, compute **avg** (salary) and find all tuples to delete
  2. Next, delete all tuples found above (without recomputing **avg** or retesting the tuples)

INSERTION插入

- Add a new tuple to *course*

```
insert into course
    values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);
```

- or equivalently

```
insert into course (course_id, title, dept_name, credits)
    values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);
```

- Add a new tuple to *student* with *tot_creds* set to null

```
insert into student
    values ('3003', 'Green', 'Finance', null);
```


INSERTION

- Add all instructors to the *student* relation with tot_creds set to 0

```
insert into student
    select ID, name, dept_name, 0
    from   instructor
```

- The **select from where** statement is evaluated fully before any of its results are inserted into the relation.

  Otherwise queries like

  insert into *table*1 **select** * **from** *table*1

  would cause problem

UPDATES更新

- Increase salaries of instructors whose salary is over $100,000 by 3%, and all others by a 5%
  - Write two **update** statements:

```
update instructor
  set salary = salary * 1.03
  where salary > 100000;
update instructor
  set salary = salary * 1.05
  where salary <= 100000;
```

  - ***The order is important***
  - Can be done better using the **case** statement (next slide)

UPDATE
- CASE STATEMENT FOR CONDITIONAL UPDATES

- Same query as before but with case statement

```
update instructor
    set salary = case
                     when salary <= 100000 then salary * 1.05
                     else salary * 1.03
                 end
```


UPDATE
- CASE STATEMENT FOR CONDITIONAL UPDATES

- The general form of the case statement is as follows

case
  when $pred_1$ **then** $result_1$
  when $pred_2$ **then** $result_2$
  . . .
  when $pred_n$ **then** $result_n$
  else $result_0$
end

UPDATES WITH SCALAR SUBQUERIES

- Recompute and update tot_creds value for all students

```
update student S
 set tot_cred = (select sum(credits)
                 from takes, course
                 where takes.course_id = course.course_id and
                   S.ID= takes.ID  and  takes.grade <> 'F' and
                   takes.grade is not null);
```

- !Sets *tot_creds* to null for students who have not taken any course

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UPDATES WITH SCALAR SUBQUERIES

- Recompute and update tot_creds value for all students, but sets tot_creds to 0 for students who have not taken any course

```
update student S
 set tot_cred = (select
      case
        when sum(credits) is not null then sum(credits)
        else 0
      end
     from takes , course
     where takes.course_id = course.course_id and
       S.ID= takes.ID  and  takes.grade <> 'F' and takes.grade is not null);
```

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CASE

```
CREATE TABLE shirts (
   name VARCHAR(40),
   size ENUM('x-small', 'small', 'medium', 'large', 'x-large')
);
INSERT INTO shirts (name, size) VALUES ('dress shirt','large'), ('t-shirt','medium'),
  ('polo shirt','small');
SELECT name, size FROM shirts WHERE size = 'medium';
+---------+--------+
| name    | size   |
+---------+--------+
| t-shirt | medium |
+---------+--------+
UPDATE shirts SET size = 'small' WHERE size = 'large';
COMMIT;
```

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SUMMARY

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